Verifying Trigonometric Identities (TE2)

Math 172 lecture notes · Section 4.4.2

Learning objective: symbolically verify identities by using various trigonometric identities.

Try each part before you open the answer. Your thinking is what makes it stick.

Problem from your notes Text from the handout Extra note (not in your handout)

Overview

FROM THE HANDOUT

Occasionally a question may ask you to "prove the identity" or "establish the identity." In these situations, you must show the algebraic manipulations that demonstrate that the left and right side of the equation are equal. You can think of a "prove the identity" problem as a simplification problem where you know the answer: you know what the end goal of the simplification should be, and just need to show the steps to get there.

To prove an identity, start with the expression on one side of the identity and manipulate it using algebra and trigonometric identities until you have simplified it to the expression on the other side of the equation. **Do not treat the identity like an equation!** The proof is establishing the two expressions are equal, so work with one side at a time rather than applying an operation simultaneously to both sides of the equation.

Toolbox

✎ EXTRA NOTE: NOT IN YOUR HANDOUT

Keep these within reach. Every step in a proof should be one of them, or plain algebra.

- **Quotient:** $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$, $\cot\theta = \dfrac{\cos\theta}{\sin\theta}$
- **Reciprocal:** $\sec\theta = \dfrac{1}{\cos\theta}$, $\csc\theta = \dfrac{1}{\sin\theta}$
- **Pythagorean:** $\sin^2\theta + \cos^2\theta = 1$ (also $1 - \sin^2\theta = \cos^2\theta$ and $1 - \cos^2\theta = \sin^2\theta$), $1 + \tan^2\theta = \sec^2\theta$, $1 + \cot^2\theta = \csc^2\theta$

An identity is true wherever both sides are defined. For example, Activity 1 only makes sense where $\cos\theta \neq 0$.

Activity 1

Activity 1 · from your notes

Follow the steps to verify the identity $\tan\theta \cdot \cos\theta = \sin\theta$.

(a) Rewrite the left-hand side so that each trigonometric function is written in terms of sine and cosine.

Answer (a)

$$\tan\theta \cdot \cos\theta = \frac{\sin\theta}{\cos\theta} \cdot \cos\theta$$

(quotient identity)

(b) Now simplify the left-hand side.

Answer (b)

The cos θ on top cancels the cos θ on the bottom (algebra).

$$\frac{\sin\theta}{\cos\theta}\cdot\cos\theta=\sin\theta$$

(c) Compare the simplified version of the left-hand side to the original right-hand side of the identity. What do you notice?

Answer (c)

The left side turned into exactly the right side, sin θ, and we never touched the right side.

They match, so the identity is verified

Domain check: tan θ needs cos θ ≠ 0, so the identity holds wherever cos θ ≠ 0.

Note

FROM THE HANDOUT

As we saw in the previous example, one method that often helps in verifying identities is to rewrite everything in terms of sine and cosine to see if one side of the equation simplifies.

Activity 2

Activity 2 · from your notes

Verify the identity: $\cos\theta+\sin\theta\tan\theta=\sec\theta$.

(a) Rewrite the left-hand side in terms of sine and cosine.

Answer (a)

$$\cos\theta+\sin\theta\cdot\frac{\sin\theta}{\cos\theta}=\cos\theta+\frac{\sin^2\theta}{\cos\theta}$$

(quotient identity, then multiply)

(b) *Find a common denominator to add the two fractions, so you have one fraction on the left-hand side.*

Answer (b)

Write $\cos\theta$ as $\frac{\cos^2\theta}{\cos\theta}$ so both pieces have denominator $\cos\theta$:

$$\frac{\cos^2\theta}{\cos\theta} + \frac{\sin^2\theta}{\cos\theta} = \frac{\cos^2\theta + \sin^2\theta}{\cos\theta}$$

(c) *Simplify the numerator of your fraction by using one of the Pythagorean Identities.*

Answer (c)

$$\frac{\cos^2\theta + \sin^2\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} \quad \text{(Pythagorean identity)}$$

$$= \sec\theta \quad \text{(reciprocal identity)}$$

Left side = sec θ, so the identity is verified

Domain check: $\tan\theta$ and $\sec\theta$ need $\cos\theta \neq 0$.

Activity 3

Activity 3 · from your notes

Verify the following identity: $\frac{\tan\theta - \cot\theta}{\sin\theta\cos\theta} = \sec^2\theta - \csc^2\theta$.

(a) *The more complex side involves a fraction that can be split up. How can we rewrite the left side so that we end up with two fractions?*

Answer (a)

One denominator under a difference splits into two fractions:

$$\frac{\tan\theta}{\sin\theta\cos\theta} - \frac{\cot\theta}{\sin\theta\cos\theta}$$

(b) Rewrite $\tan\theta$ and $\cot\theta$ in terms of sine and cosine.

Answer (b)

$$\frac{\frac{\sin\theta}{\cos\theta}}{\sin\theta\cos\theta} - \frac{\frac{\cos\theta}{\sin\theta}}{\sin\theta\cos\theta}$$

(quotient identities)

(c) Simplify each complex fraction to verify that the left-hand side is equal to $\sec^2\theta - \csc^2\theta$.

Hint: Dividing by $(\sin\theta\cos\theta)$ is the same as multiplying the denominator by $\sin\theta\cos\theta$.

Answer (c)

First fraction: $\frac{\sin\theta}{\cos\theta} \div (\sin\theta\cos\theta) = \frac{\sin\theta}{\cos\theta \cdot \sin\theta\cos\theta} = \frac{1}{\cos^2\theta}$

$= \sec^2\theta$

Second fraction: $\frac{\cos\theta}{\sin\theta} \div (\sin\theta\cos\theta) = \frac{\cos\theta}{\sin\theta \cdot \sin\theta\cos\theta} = \frac{1}{\sin^2\theta} = \csc^2\theta$

Left side = $\sec^2\theta - \csc^2\theta$, so the identity is verified

⚠ **Watch out:** Keep the minus sign in front of the second fraction the whole way through.

Domain check: both sides need $\sin\theta \neq 0$ and $\cos\theta \neq 0$, that is, $\theta \neq k \cdot \pi/2$, where k is any integer.

Note

FROM THE HANDOUT

Starting with the more complex side can sometimes make the simplification easier.

Activity 4

Activity 4 · from your notes

Refer to previous work to help you verify the identity:

$\frac{\csc\theta - \sin\theta}{\sin\theta\csc\theta} = \csc\theta - \sin\theta$. **Justify each step by listing what**

you used, such as algebra, Pythagorean identity, reciprocal identities, etc.

Try it, then check.

> **Hint:** Activity 3 split a fraction with a difference on top. Try that here.

Answer

$$\frac{\csc\theta - \sin\theta}{\sin\theta\csc\theta} \quad \text{(start with the left side)}$$

$$= \frac{\csc\theta}{\sin\theta\csc\theta} - \frac{\sin\theta}{\sin\theta\csc\theta} \quad \text{(algebra: split the fraction)}$$

$$= \frac{1}{\sin\theta} - \frac{1}{\csc\theta} \quad \text{(algebra: cancel common factors)}$$

$$= \csc\theta - \sin\theta \quad \text{(reciprocal identities)}$$

Left side = csc θ − sin θ, so the identity is verified

> **Domain check:** $\csc\theta$ needs $\sin\theta \neq 0$ ($\theta \neq k\pi$, where k is any integer). Then $\sin\theta\csc\theta = 1$, so the denominator is never 0.

Activity 5

Activity 5 · from your notes

Verify the following identity: $\frac{\cos^2\theta}{1+\sin\theta} = 1 - \sin\theta$.

(a) *The left side is more complicated, so start there. The right side only involves sine. Which identity could help us rewrite $\cos^2\theta$ in terms of sine?*

Answer (a)

$\sin^2\theta + \cos^2\theta = 1$, so $\cos^2\theta = 1 - \sin^2\theta$

(b) *Rewrite the numerator of the left-hand side as a function of sine.*

Answer (b)

$$\frac{\cos^2\theta}{1+\sin\theta} = \frac{1-\sin^2\theta}{1+\sin\theta}$$

(c) Take a look at the numerator you now have. How can we factor it?

Answer (c)

(algebra) It's a difference of squares, $a^2 - b^2 = (a - b)(a + b)$, with $a = 1$ and $b = \sin\theta$.

$$1 - \sin^2\theta = (1 - \sin\theta)(1 + \sin\theta)$$

(d) Simplify by canceling out terms.

Answer (d)

$\frac{(1 - \sin\theta)(1 + \sin\theta)}{1 + \sin\theta}$: the $(1 + \sin\theta)$ cancels.

Left side = $1 - \sin\theta$, so the identity is verified

Domain check: canceling $(1 + \sin\theta)$ needs $\sin\theta \neq -1$. At $\theta = 3\pi/2$ the left side is $0/0$, undefined, even though the right side equals 2.

Note

FROM THE HANDOUT

As we saw in the previous example, knowing how to factor can be very useful when simplifying trigonometric identities.

Note

FROM THE HANDOUT

Using the property of conjugates is sometimes helpful in simplifying trigonometric identities. For an expression like $a + b$, the conjugate would be $a - b$. When you multiply conjugates, you often get a more useful expression. Sometimes multiplying by the conjugate will simplify an expression and help in verifying the given identity. Let's try this method in the next activity.

Activity 6

Activity 6 · from your notes

Verify the following identity: $\frac{\cos\theta}{1 - \sin\theta} = \frac{1 + \sin\theta}{\cos\theta}$.

(a) Start with the left-hand side. Multiply it by the conjugate of $1 - \sin\theta$.

Hint: When multiplying by the conjugate, do not distribute the $\cos\theta$ in the numerator.

Answer (a)

Multiply by $\frac{1+\sin\theta}{1+\sin\theta}$, which equals 1, so the value doesn't change:

$$\frac{\cos\theta\,(1+\sin\theta)}{(1-\sin\theta)(1+\sin\theta)} = \frac{\cos\theta\,(1+\sin\theta)}{1-\sin^2\theta}$$

(b) Rewrite the denominator of your fraction in part (a) so that it is written in terms of cosine.

Hint: Use a Pythagorean Identity.

Answer (b)

$$\frac{\cos\theta\,(1+\sin\theta)}{\cos^2\theta}$$

(Pythagorean identity: $1-\sin^2\theta = \cos^2\theta$)

(c) Simplify your expression from part (b) to get the right-hand side.

Answer (c)

One $\cos\theta$ cancels from top and bottom (algebra).

$$\frac{\cos\theta\,(1+\sin\theta)}{\cos^2\theta} = \frac{1+\sin\theta}{\cos\theta}, \text{ so the identity is verified}$$

Domain check: multiplying by $(1+\sin\theta)/(1+\sin\theta)$ needs $\sin\theta \neq -1$, so that the factor is nonzero and really equals 1; and the left side needs $\sin\theta \neq 1$. Together: $\cos\theta \neq 0$, the same restriction as the right side.

Note

FROM THE HANDOUT

As we've seen from the activities from this section, there are some basic tools that can be helpful when verifying trigonometric identities. Here are some suggestions as you continue to work through these types of problems.

Work on one side of the identity. It is usually better to start with the more complex side, as it is easier to simplify than to build. Look for opportunities to

- Multiply expressions out and combine like terms.
- Factor expressions in a fraction and cancel common factors.
- Split apart fractions.
- Rewrite multiple fractions using a common denominator, and combine the fractions into a single fraction.
- Simplify two-term denominators by substituting with a Pythagorean identity.
- Simplify two-term denominators by multiplying numerator and denominator by the conjugate of the binomial denominator.

Observe which functions are in the final expression, and look for opportunities to use identities that would lead to those functions. If all else fails, try rewriting all terms to sines and cosines.

Activity 7

Activity 7 · from your notes

Use the tools you have learned in this section to verify the identity: $2\tan\theta\sec\theta = \dfrac{2\sin\theta}{1-\sin^2\theta}$.

Try it, then check.

> **Hint:** Which side looks more complicated? Which strategy from the Note handles a two-term denominator?

Answer

The right side has the two-term denominator, so start there.

$\dfrac{2\sin\theta}{1-\sin^2\theta}$ (start with the right side)

$= \dfrac{2\sin\theta}{\cos^2\theta}$ (Pythagorean identity: $1-\sin^2\theta = \cos^2\theta$)

$= 2 \cdot \dfrac{\sin\theta}{\cos\theta} \cdot \dfrac{1}{\cos\theta}$ (algebra: split $\cos^2\theta$ into $\cos\theta \cdot \cos\theta$)

$= 2\tan\theta\sec\theta$ (quotient and reciprocal identities)

Right side = $2\tan\theta\sec\theta$, so the identity is verified

Domain check: both sides need $\cos\theta \neq 0$ (equivalently $1-\sin^2\theta \neq 0$).

Practice Example 8

Practice Example 8 · from your notes

Verify the identity $\sec(x) - \sin(x)\tan(x) = \cos(x)$ by simplifying the left side.

Try it, then check.

Answer

$\sec x - \sin x \tan x$ *(start with the left side)*

$= \dfrac{1}{\cos x} - \sin x \cdot \dfrac{\sin x}{\cos x}$ *(reciprocal and quotient identities)*

$= \dfrac{1 - \sin^2 x}{\cos x}$ *(algebra: common denominator)*

$= \dfrac{\cos^2 x}{\cos x}$ *(Pythagorean identity)*

$= \cos x$ *(algebra: cancel)*

Left side = cos x, so the identity is verified

Domain check: $\sec x$ and $\tan x$ need $\cos x \neq 0$. $\cos x$ is defined everywhere, but the identity only holds where the left side is defined.

Practice Example 9

Practice Example 9 · from your notes

Verify the identity $\dfrac{\sin(x)}{1 - \cos(x)} = \dfrac{1 + \cos(x)}{\sin(x)}$ by multiplying the left side by a well-chosen form of 1.

Try it, then check.

Hint: Compare with Activity 6. What is the conjugate of $1 - \cos x$?

Answer

$\dfrac{\sin x}{1 - \cos x}$ *(start with the left side)*

$= \dfrac{\sin x}{1 - \cos x} \cdot \dfrac{1 + \cos x}{1 + \cos x}$ *(multiply by 1, using the conjugate)*

$= \dfrac{\sin x\,(1 + \cos x)}{1 - \cos^2 x}$ *(algebra: difference of squares)*

$= \dfrac{\sin x\,(1 + \cos x)}{\sin^2 x}$ *(Pythagorean identity)*

$= \dfrac{1 + \cos x}{\sin x}$ *(algebra: cancel one sin x)*

Left side = the right side, so the identity is verified

Domain check: both sides need $\sin x \neq 0$, that is, $x \neq k\pi$, where k is any integer. That also guarantees $1 - \cos x \neq 0$ and $1 + \cos x \neq 0$, so the factor $(1 + \cos x)/(1 + \cos x)$ we multiplied by really equals 1.

Summary

FROM THE HANDOUT

Write 2–3 sentences summarizing the key concepts of today's lecture.

Accessibility & help

These notes are built to work with screen readers, keyboard-only use, zoom and phone screens. Every answer stays hidden until you open it, and **Switch to standard font** at the top changes the handwriting to a plain font.

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